



PAPER ID-310442

Printed Page: 1 of 2
Subject Code: RCAI703

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MCA-INT
(SEM VII) THEORY EXAMINATION 2024-25
COMPUTER BASED OPTIMIZATION TECHNIQUES

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A1. Attempt *all* questions in brief.

2 x 07 = 14

Q no.	Question	CO	Level
a.	Define Operations Research (OR).	1	1
b.	Explain the purpose of the Simplex Method.	1	1
c.	What is degeneracy in a Transportation Problem?	2	1
d.	Define an Assignment Problem (AP).	2	1
e.	Define the Branch and Bound Method.	3	1
f.	What is an inventory model?	4	1
g.	What is Bellman's Principle of Optimality?	5	1

SECTION B2. Attempt *any three* of the following:

07 x 3 = 07

a.	Compare the Two-Phase Method and Big-M Method for solving LPPs.	1	2
b.	Explain the Transportation Problem (TP) and discuss its applications in operations management.	2	2
c.	Explain the concept of replacement problems with examples of capital equipment replacement.	3	2
d.	Discuss the various types of costs considered in inventory models and their significance.	4	2
e.	Solve a linear programming problem using dynamic programming and explain the process in detail.	5	2

SECTION C3. Attempt *any one* part of the following:

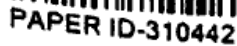
07 x 1 = 07

a.	Explain the nature, meaning, and main characteristics of Operations Research (OR) with suitable examples.	1	2
b.	Solve a Linear Programming Problem (LPP) using the graphical method with proper steps.	1	2

4. Attempt *any one* part of the following:

07 x 1 = 07

a.	Use Matrix minima, Column minima and Row minima method to find initial basic feasible solution to the transportation problem:	2	2																														
<table><tr><td></td><td>D1</td><td>D2</td><td>D3</td><td>D4</td><td>Supply</td></tr><tr><td>S1</td><td>19</td><td>30</td><td>50</td><td>10</td><td>7</td></tr><tr><td>S2</td><td>70</td><td>30</td><td>40</td><td>60</td><td>9</td></tr><tr><td>S3</td><td>40</td><td>8</td><td>70</td><td>20</td><td>18</td></tr><tr><td>Demand</td><td>5</td><td>8</td><td>7</td><td>14</td><td>34</td></tr></table>			D1	D2	D3	D4	Supply	S1	19	30	50	10	7	S2	70	30	40	60	9	S3	40	8	70	20	18	Demand	5	8	7	14	34		
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b.	Explain the steps involved in solving the Hungarian Method for an Assignment Problem (AP) with a numerical example.	2	2																														



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| 5. Attempt any <i>one</i> part of the following: | | 07 x 1 = 07 | |
| a. | Differentiate between pure and mixed integer linear programming problems with examples. | 3 | 2 |
| b. | Discuss the Cutting Plane Method in detail with an example to illustrate its working. | 3 | 2 |

6. Attempt any <i>one</i> part of the following:		07 x 1 = 07	
a.	Explain deterministic inventory models and their applications in inventory management.	4	2
b.	Describe the stochastic inventory models and their applications in decision-making.	4	2

7. Attempt any <i>one</i> part of the following:		07 x 1 = 07	
a.	Discuss the concept of a multistage decision problem and how dynamic programming is used to solve it.	5	2
b.	Explain the role of dynamic programming in decision-making for optimization problems.	5	2